On Self-Conjugate $(t, t+1, \ldots, t+p)$-Core Partitions into Distinct Parts

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Abstract. Simultaneous core partitions have been extensively studied over the past two decades. Significant attention has been paid to simultaneous core partitions into distinct parts since Amdeberhan's conjectures on explicit formulas for the number, the largest size, and the average size of $(t, t+1)$-core partitions into distinct parts in 2015. These conjectures were subsequently proved and generalized by Straub, Zaleski, Xiong, and Paramonov, among others. Recently, the authors introduced a proper self-conjugate partition analog of $(t, t+1)$-core partitions into distinct parts and established formulas for their enumeration, average size, and largest size. In this paper, we extend this framework to self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts and derive explicit formulas for their largest size. We also obtain generating functions for the number of such partitions when $p = 2$ and $p = 3$. Additionally, we provide numerical data computed via Python for $p \leq 10$ and $t \leq 18$.

Keywords. Integer partition, self-conjugate partition, hook length, core partition, average size, largest size.

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1 Introduction

For a positive integer n, $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_m)$ is called a *partition* of n if $\lambda_1 + \lambda_2 + \cdots + \lambda_m = n$, where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m \geq 1$ are positive integers. We denote this by $\lambda \vdash n$. For $n = 0$, the empty sequence $\varnothing = ()$ is considered a partition of 0. A partition λ is commonly represented by its *Young diagram*, consisting of left-justified rows with λ_i boxes in the i-th row. The *conjugate partition* of λ is obtained by reflecting its Young diagram along the main diagonal. The partition λ is said to be self-conjugate if it is identical to its conjugate. The *hook length* of the box $\square = (i, j)$ in the i-th

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row and the j-th column, denoted by $h(i,j)$, is defined as the number of boxes to its right and below it, plus the box itself. For example, the Young diagram of the self-conjugate partition $(8,6,4,3,2,2,1,1)$ and its hook lengths are shown in Fig. 1.

15	12	9	7	5	4	2	1
12	9	6	4	2	1		
9	6	3	1				
7	4	1					
5	2						
4	1						
2							
1							

Fig. 1: The Young diagram and hook lengths of the self-conjugate partition $(8,6,4,3,2,2,1,1)$.

For a positive integer t, a partition is called a *t-core partition* if none of its hook lengths is divisible by t. Let t_i $(1 \le i \le k)$ be positive integers. We say that λ is a $(t_1, t_2, \ldots, t_k)$-core partition if it is simultaneously a t_1-core, a t_2-core, ..., and a t_k-core partition. For example, as shown in Fig. 1, the partition $(8,6,4,3,2,2,1,1)$ is a $(10,11)$-core partition. It is well known that the number of $(t_1, t_2, \ldots, t_k)$-core partitions is finite if and only if $\gcd(t_1, t_2, \ldots, t_k) = 1$ (see, e.g., [29]).

The study of simultaneous core partitions can be traced back to the work of Anderson [3] and has become a research focus in enumerative combinatorics over the past twenty years. The results on the number, largest size, and average size of such partitions can be found in [2, 3, 5, 8–21, 25, 28, 30–32, 34, 36–40]. For example, Anderson [3] showed that the number of (s,t)-core partitions equals the rational Catalan number $\frac{1}{s+t}\binom{s+t}{s}$ when s and t are coprime. Aukerman, Kane, and Sze [6] conjectured that the largest size of an (s,t)-core partition equals $\frac{(s^2-1)(t^2-1)}{24}$, which was later proved by Olsson and Stanton [23]. In 2014, Armstrong, Hanusa, and Jones [5] conjectured that the average size of (s,t)-core partitions is equal to $\frac{(s+t-1)(s-1)(t-1)}{24}$. This conjecture was later proved independently by Johnson [16] and Wang [28].

Analogous results have also been obtained for self-conjugate core partitions. Ford, Mai, and Sze [14] established an analogue of Anderson's result. They characterized the hook lengths of the diagonal boxes in self-conjugate (s,t)-core partitions and showed that the number of self-conjugate (s,t)-core partitions is equal to $\binom{\lfloor \frac{s}{2} \rfloor + \lfloor \frac{t}{2} \rfloor}{\lfloor \frac{s}{2} \rfloor}$ when s and t are coprime. Furthermore, Olsson and Stanton [23] proved

that the (s,t)-core partition of the largest size is necessarily self-conjugate, hence the largest size in this setting is also equal to $\frac{(s^2-1)(t^2-1)}{24}$. The conjecture proposed by Armstrong, Hanusa, and Jones [5] in 2014 stated that the average size of such partitions is $\frac{(s+t-1)(s-1)(t-1)}{24}$. This result was established by Chen, Huang, and Wang [8] in 2016.

Significant attention has been devoted to simultaneous core partitions into distinct parts since Amdeberhan's conjectures [1] in 2015. He conjectured explicit formulas for the number, the largest size, and the average size of $(t,t+1)$-core partitions into distinct parts, these conjectures were subsequently proved and generalized by Baek, Nam, and Yu [7], Nath and Sellers [22], Paramonov [24], Straub [26], the first author [31], Yan, Qin, Jin, and Zhou [35], Zaleski [38, 39], Zaleski and Zeilberger [40], etc. In particular, Amdeberhan [1] conjectured that the number of $(t,t+1)$-core partitions into distinct parts equals the $(t+1)$-st Fibonacci number. Let F_t denote the t-th Fibonacci number, defined by

$$F_t = F_{t-1} + F_{t-2}, \quad t \geq 3,$$

with initial values $F_1 = F_2 = 1$. In [33], the authors obtained analogous results for self-conjugate $(t,t+1)$-core partitions into distinct parts. We begin by recalling the definition of the Durfee square. For a partition λ, its Durfee square has size s if s is the largest integer such that λ has at least s parts, each of which is at least s (see [4]). Based on the concept of the Durfee square, we introduce the following definition.

Definition 1.1 ([33]) *Let t_i $(1 \leq i \leq k)$ be positive integers. A self-conjugate $(t_1,t_2,\ldots,t_k)$-core partition λ is called a self-conjugate $(t_1,t_2,\ldots,t_k)$-core partition into distinct parts if its first $s(\lambda)$ parts are distinct, where $s(\lambda)$ denotes the size of the Durfee square of λ.*

For example, the partition $\lambda = (8,6,4,3,2,2,1,1)$ is a self-conjugate $(10,11)$-core partition into distinct parts since $s(\lambda) = 3$ and its first three parts 8, 6, and 4 are distinct. In [33], we determined the number, the average size, and the largest size for self-conjugate $(t,t+1)$-core partitions into distinct parts. More precisely, we showed that the number G_t of such partitions satisfies the recursion

$$G_t = G_{t-1} + G_{t-3}, \quad t \geq 4,$$

with the initial values $G_1 = 1$, $G_2 = 2$, and $G_3 = 3$. The sequence $(G_t)_{t\geq 1}$ has appeared in the literature with different initial conditions; see Sequence $A000930$ of the On-line Encyclopedia of Integer Sequences (OEIS) [27]. This recurrence implies that the enumeration of $(t,t+1)$-core partitions into distinct parts and that of their

self-conjugate counterparts follows similar recursive patterns. Furthermore, their largest and average size also exhibit similar behaviors. Consequently, self-conjugate $(t, t+1)$-core partitions into distinct parts defined in [33] can be viewed as a natural self-conjugate counterpart to $(t, t+1)$-core partitions into distinct parts.

In this paper, building upon previous results in [33], we study the number and the largest size of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts. For positive integers t and p, we denote $\mathcal{DS}_p(t)$ by the set of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts, for notational convenience. Note that $\mathcal{DS}_p(t)$ contains the empty partition $\varnothing$. In Table 1, we list the number of partitions in $\mathcal{DS}_2(t)$ and the explicit sets of these partitions for $1 \le t \le 9$.

Table 1: The number of partitions in $\mathcal{DS}_2(t)$ and the set of such partitions.

t	*The set* $(\mathcal{DS}_2(t))$	*The number* $(f(t))$
1	$\{\varnothing\}$	1
2	$\{\varnothing, (1)\}$	2
3	$\{\varnothing, (1)\}$	2
4	$\{\varnothing, (1), (2,1), (4,1,1,1)\}$	4
5	$\{\varnothing, (1), (2,1), (5,1,1,1,1)\}$	4
6	$\{\varnothing, (1), (2,1), (3,1,1), (3,2,1),$ $(5,1,1,1,1), (5,2,1,1,1), (6,1,1,1,1,1)\}$	8
7	$\{\varnothing, (1), (2,1), (3,1,1), (3,2,1),$ $(6,1,1,1,1,1), (6,2,1,1,1,1), (7,1,1,1,1,1,1)\}$	8
8	$\{\varnothing, (1), (2,1), (3,1,1), (3,2,1), (4,1,1,1),$ $(4,2,1,1), (4,3,2,1), (6,1,1,1,1,1), (6,2,1,1,1,1),$ $(6,3,2,1,1,1), (7,1,1,1,1,1,1), (7,2,1,1,1,1,1),$ $(8,1,1,1,1,1,1,1), (8,5,2,2,2,1,1,1), (8,7,2,2,2,2,2,1)\}$	16
9	$\{\varnothing, (1), (2,1), (3,1,1), (3,2,1), (4,1,1,1),$ $(4,2,1,1), (4,3,2,1), (7,1,1,1,1,1,1), (7,2,1,1,1,1,1),$ $(7,3,2,1,1,1,1), (8,1,1,1,1,1,1,1), (8,2,1,1,1,1,1,1),$ $(9,1,1,1,1,1,1,1,1), (9,5,2,2,2,1,1,1,1), (9,8,2,2,2,2,2,2,1)\}$	16

We present the following enumeration results for self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts for $p = 2$ and $p = 3$.

Theorem 1.2 *Suppose that t is a positive integer. Let $f(t)$ denote the number of partitions in $\mathcal{DS}_2(t)$ with $f(0) = 1$. Then, for $t \ge 1$, $f(t) = 2^{\lfloor \frac{t}{2} \rfloor}$ and the ordinary generating function of $f(t)$ is*

$$\sum_{t \ge 0} f(t) x^t = \frac{1+x}{1-2x^2}. \tag{1.1}$$

Theorem 1.3 *Suppose that t is a positive integer. Let $g(t)$ denote the number of partitions in $\mathcal{DS}_3(t)$ with $g(0) = 1$. Then $g(1) = 1$, $g(2) = g(3) = 2$, $g(4) = 3$, and for $t \geq 5$, we have $g(t) = g(t-1) + g(t-4)$. Thus, the ordinary generating function of $g(t)$ is*

$$\sum_{t\geq 0} g(t)x^t = \frac{1+x^2}{1-x-x^4}. \tag{1.2}$$

Remark 1.4 *The sequence $(g(t))_{t\geq 0}$ has appeared previously, with different initial conditions; see Sequence A103632 of OEIS [27].*

We also derive results on the largest size of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts for the general case $p \geq 2$.

Theorem 1.5 *Suppose that $p \geq 2$ and $t \geq 1$ are positive integers. Let $L_p(t)$ denote the largest size of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts. For a statement S, let $\delta(S) = 1$ if S is true and $\delta(S) = 0$ otherwise. We then have the following results.*

(i) If $t - p \equiv 0 \pmod 4$, then $L_p(t)$ equals

$$\max\{\frac{(3t+p+2)(t-p)}{8}\delta(t-p>1),\ \frac{t(t+2)}{8}\delta(t \text{ is even}),\ \frac{(t-1)(t+1)}{8}\delta(t \text{ is odd})\}.$$

(ii) If $t - p \equiv 1 \pmod 4$, then $L_p(t)$ equals

$$\max\{\frac{(3t+p+3)(t-p-1)}{8}\delta(t-p>1),\ \frac{t(t+2)}{8}\delta(t \text{ is even}),\ \frac{(t-1)(t+1)}{8}\delta(t \text{ is odd})\}.$$

(iii) If $t - p \equiv 2 \pmod 4$, then $L_p(t)$ equals

$$\max\{\frac{(3t+p)(t-p+2)}{8}\delta(t-p>1),\ \frac{t(t+2)}{8}\delta(t \text{ is even}),\ \frac{(t-1)(t+1)}{8}\delta(t \text{ is odd})\}.$$

(iv) If $t - p \equiv 3 \pmod 4$, then $L_p(t)$ equals

$$\max\{\frac{(3t+p+1)(t-p+1)}{8}\delta(t-p>1),\ \frac{t(t+2)}{8}\delta(t \text{ is even}),\ \frac{(t-1)(t+1)}{8}\delta(t \text{ is odd})\}.$$

Moreover, the number of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts with the largest size is at most 2.

By Theorem 1.5, we obtain the following corollary by straightforward calculations for $p = 2$ and $p = 3$.

Corollary 1.6 *Let t be a positive integer. For partitions in $\mathcal{DS}_2(t)$, we have $L_2(1) = 0$, $L_2(2) = 1$, $L_2(3) = 1$, and for $t \geq 4$,*

$$L_2(t) = \begin{cases} \frac{(3t+2)t}{8} & \text{if } t \equiv 0 \pmod 4, \\ \frac{(3t+3)(t-1)}{8} & \text{if } t \equiv 1 \pmod 4, \\ \frac{(3t+4)(t-2)}{8} & \text{if } t \equiv 2 \pmod 4, \\ \frac{(3t+5)(t-3)}{8} & \text{if } t \equiv 3 \pmod 4. \end{cases}$$

Corollary 1.7 *Let t be a positive integer. For partitions in $\mathcal{DS}_3(t)$, we have $L_3(1) = 0$, $L_3(2) = 1$, $L_3(3) = 1$, $L_3(4) = 3$, and for $t \geq 5$,*

$$L_3(t) = \begin{cases} \frac{(3t+6)(t-4)}{8} & \text{if } t \equiv 0 \pmod 4, \\ \frac{(3t+3)(t-1)}{8} & \text{if } t \equiv 1 \pmod 4, \\ \frac{(3t+4)(t-2)}{8} & \text{if } t \equiv 2 \pmod 4, \\ \frac{(3t+5)(t-3)}{8} & \text{if } t \equiv 3 \pmod 4. \end{cases}$$

We will use the following notation and terminology. Let $\mathbb{N}_s = \{1, 2, \ldots, s\}$ and $\mathbb{N}_s^0 = \mathbb{N}_s \cup \{0\}$. For a self-conjugate partition λ, define the set of main diagonal hook lengths of λ (hook lengths of the boxes in the main diagonal of the Young diagram of λ) to be $MD(\lambda) = \{h_1, h_2, \ldots, h_k\}$, where $h_1 > h_2 > \cdots > h_k$ and $k = s(\lambda)$ is the number of main diagonal hook lengths of λ, which is also denoted by $|MD(\lambda)|$. Note that when λ is self-conjugate, the elements of $MD(\lambda)$ are distinct odd integers (by convention $MD(\varnothing) = \varnothing$). For example, if $\lambda = (8, 6, 4, 3, 2, 2, 1, 1)$, then $MD(\lambda) = \{15, 9, 3\}$. Let $d(\lambda) = (d_1, d_2, \ldots, d_{k-1})$ denote the sequence of consecutive differences of these main diagonal hook lengths, where $d_i = h_i - h_{i+1}$ for $i \in \mathbb{N}_{k-1}$. For the example above, we have $d(\lambda) = (d_1, d_2) = (15 - 9, 9 - 3) = (6, 6)$. For a positive integer m, we say $d(\lambda) \geq m$ if $d_i \geq m$ for all $1 \leq i \leq k - 1$. Under this convention, one easily checks that a self-conjugate core partition λ is a self-conjugate core partition into distinct parts if and only if $d(\lambda) \geq 4$ (when $k = 1$, $d(\lambda)$ is the empty sequence, so $d(\lambda) \geq 4$ holds automatically). In Table 2, we list the largest partition sizes in $\mathcal{DS}_3(t)$ and their $MD(\lambda)$ for $2 \leq t \leq 19$.

Table 2: Largest size of partitions in $\mathcal{DS}_3(t)$ and their $MD(\lambda)$.

t	$MD(\lambda)$	*Largest size*	t	$MD(\lambda)$	*Largest size*
2	$\{1\}$	1	11	$\{21,17\}$	38
3	$\{1\}$	1	12	$\{23,19\}$	42
4	$\{3\}$	3	13	$\{25,21,17\}$	63
5	$\{9\}$	9	14	$\{27,23,19\}$	69
6	$\{11\}$	11	15	$\{29,25,21\}$	75
7	$\{13\}$	13	16	$\{31,27,23\}$	81
8	$\{15\}$	15	17	$\{33,29,25,21\}$	108
9	$\{17,13\}$	30	18	$\{35,31,27,23\}$	116
10	$\{19,15\}$	34	19	$\{37,33,29,25\}$	124

The remainder of the paper is organized as follows. In section 2, we characterize the main diagonal hook lengths of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts. Using these structural properties and a suitable bijection, we prove Theorems 1.2 and 1.3. In section 3, we describe the main diagonal hook lengths of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts of the largest size, thereby establishing Theorem 1.5.

2 The number of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts for $p = 2, 3$

We briefly review the structural properties of self-conjugate t-core partitions. Ford, Mai, and Sze [14] characterized the set $MD(\lambda)$ for a self-conjugate core partition.

Theorem 2.1 ([14, Proposition 3]) *Let λ be a self-conjugate partition. Then λ is a t-core partition if and only if the following conditions hold.*

(i) *If $h \in MD(\lambda)$ and $h > 2t$, then $h - 2t \in MD(\lambda)$.*

(ii) *If $h_1, h_2 \in MD(\lambda)$, then $h_1 + h_2$ is not a multiple of $2t$.*

The definition of the main diagonal hook length lets us obtain the following result directly.

Lemma 2.2 *The size of a self-conjugate partition is equal to the sum of its main diagonal hook lengths, that is, if λ is a self-conjugate partition, then*

$$|\lambda| = \sum_{x \in MD(\lambda)} x.$$

In [33], relying on Theorem 2.1, we characterized the set $MD(\lambda)$ for a self-conjugate $(t, t+1)$-core partition into distinct parts and established the following lemma.

Lemma 2.3 ([33, Lemma 2.3]) *Let t be a positive integer and λ be a self-conjugate $(t, t+1)$-core partition into distinct parts. If $h \in MD(\lambda)$, then $h \leq 2t-1$.*

For positive integers t and p, any self-conjugate $(t, t+1, \ldots, t+p)$-core partition into distinct parts is necessarily a self-conjugate $(t, t+1)$-core partition into distinct parts. Thus Lemma 2.3 implies the following result directly.

Lemma 2.4 *Let t and p be positive integers and λ be a self-conjugate $(t, t+1, \ldots, t+p)$-core partition into distinct parts. If $h \in MD(\lambda)$, then $h \leq 2t-1$.*

For integers $t \geq 1$ and $p \geq 2$, let

$$\mathcal{H}_t^p = \{1, 3, 5, \ldots, 2t-1\} \backslash \{t, t+1, \ldots, t+p\}.$$

For example, if $t = 12$ and $p = 2$, then we have

$$\mathcal{H}_{12}^2 = \{1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23\}.$$

For integers $a \leq b$, we define $[a, b] = \{a, a+1, \ldots, b\}$. We define two subsets of $\mathcal{H}_t^p$:

$$\mathcal{LH}_t^p = \mathcal{H}_t^p \cap [t+p+1, 2t-1] \quad \text{and} \quad \mathcal{RH}_t^p = \mathcal{H}_t^p \cap [1, t-1].$$

Then $\mathcal{H}_t^p = \mathcal{LH}_t^p \cup \mathcal{RH}_t^p$. The set $\mathcal{H}_t^p$ can be represented as a graph, where the vertices in the top row correspond to the elements of $\mathcal{LH}_t^p$ and the vertices in the bottom row correspond to the elements of $\mathcal{RH}_t^p$. An edge is drawn between two vertices if the sum of their values equals $2(t+r)$ for some $r \in \mathbb{N}_p^0$. In the above example, we have

$$\mathcal{LH}_{12}^2 = \{15, 17, 19, 21, 23\} \quad \text{and} \quad \mathcal{RH}_{12}^2 = \{1, 3, 5, 7, 9, 11\}.$$

Fig. 2 illustrates the representation of $\mathcal{H}_{12}^2$, where an edge connects two elements if and only if their sum equals 24, 26, or 28.

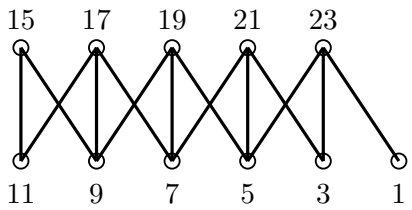


Fig. 2: The representation of $\mathcal{LH}_{12}^2$ and $\mathcal{RH}_{12}^2$.

By Theorem 2.1 and Lemma 2.4, we easily obtain the following result.

Theorem 2.5 *Let t and p be positive integers. Then λ is a partition in $\mathcal{DS}_p(t)$ if and only if $MD(\lambda)$ satisfies the following properties.*

(i) $MD(\lambda) \subseteq \mathcal{H}_t^p$.

(ii) *If $h_1, h_2 \in MD(\lambda)$ and $h_1 > h_2$, then $h_1 - h_2 \geq 4$.*

(iii) *If $h_1, h_2 \in MD(\lambda)$, then $h_1 + h_2$ is not equal to $2(t + r)$ for each $r \in \mathbb{N}_p^0$.*

2.1 Self-conjugate $(t, t+1, t+2)$-core partitions into distinct parts

We now study self-conjugate $(t, t+1, t+2)$-core partitions into distinct parts. The following theorem is an immediate consequence of Theorem 2.5.

Theorem 2.6 *Let t be a positive integer. Then λ is a partition in $\mathcal{DS}_2(t)$ if and only if $MD(\lambda)$ satisfies the following properties.*

(i) $MD(\lambda) \subseteq \mathcal{H}_t^2$.

(ii) *If $h_1, h_2 \in MD(\lambda)$ and $h_1 > h_2$, then $h_1 - h_2 \geq 4$.*

(iii) *If $h_1, h_2 \in MD(\lambda)$, then $h_1 + h_2$ is not equal to $2t$, $2t+2$ or $2t+4$.*

We need the following lemma before we prove Theorem 1.2. Recall that $f(t)$ denotes the number of partitions in $\mathcal{DS}_2(t)$ for $t \geq 1$.

Lemma 2.7 *Let $t \geq 1$ be an even integer. Then the number of partitions in $\mathcal{DS}_2(t)$ equals that in $\mathcal{DS}_2(t+1)$, i.e., if $t = 2k$, then $f(2k) = f(2k+1)$.*

Proof. Suppose that $t = 2k$ for some positive integer k. Let $\lambda \in \mathcal{DS}_2(2k)$ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$, where $x_i > x_{i+1}$ for $i \in \mathbb{N}_{m-1}$. Clearly $MD(\lambda) \subseteq \mathcal{H}_{2k}^2$. Define a map ψ_{2k}: $\mathcal{DS}_2(2k) \to \mathcal{DS}_2(2k+1)$ as follows. If $\lambda = \varnothing$, then $\psi_{2k}(\lambda) = \varnothing$. If $\lambda \neq \varnothing$, then we define $\psi_{2k}(\lambda)$ to be the partition μ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, where

$$y_i = \begin{cases} x_i & \text{if } x_i \in \mathcal{RH}_{2k}^2, \\ x_i + 2 & \text{if } x_i \in \mathcal{LH}_{2k}^2. \end{cases}$$

Then we deduce that $MD(\mu) \subseteq \mathcal{H}^2_{2k+1}$. Moreover, we deduce that $MD(\mu)$ satisfies the three properties in Theorem 2.6. This means that $\mu \in \mathcal{DS}_2(2k+1)$. Hence, the map ψ_{2k} is well-defined.

For example, if $t = 12$ and $\lambda = (10, 7, 4, 3, 2, 2, 2, 1, 1, 1) \in \mathcal{DS}_2(12)$, then we have $MD(\lambda) = \{19, 11, 3\}$, where $19 \in \mathcal{LH}^2_{12}$ and $11, 3 \in \mathcal{RH}^2_{12}$. Applying the map ψ_{12} to λ, we obtain $MD(\psi_{12}(\lambda)) = \{21, 11, 3\}$, thus $\psi_{12}(\lambda) = (11, 7, 4, 3, 2, 2, 2, 1, 1, 1, 1) \in \mathcal{DS}_2(13)$.

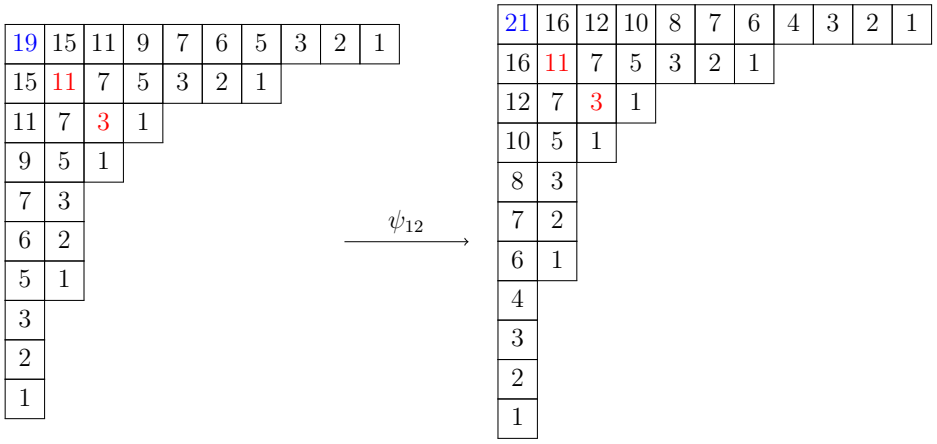


Fig. 3: An example of $\psi_{12}(\lambda)$.

Conversely, given a partition $\mu \in \mathcal{DS}_2(2k+1)$, we can recover the partition $\lambda \in \mathcal{DS}_2(2k)$ as follows. If $\mu = \varnothing$, then $\lambda = \varnothing$. If $\mu \neq \varnothing$ and $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, then we define the partition λ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$, where

$$x_i = \begin{cases} y_i & \text{if } y_i \in \mathcal{RH}^2_{2k+1}, \\ y_i - 2 & \text{if } y_i \in \mathcal{LH}^2_{2k+1}. \end{cases}$$

Therefore, the construction of the map ψ_{2k} is reversible, and hence it is a bijection. Thus $f(2k) = f(2k+1)$, completing the proof. ∎

The following result plays an essential role in the proof of Theorem 1.2.

Theorem 2.8 *Let $t \geq 1$ and $f(t)$ denote the number of partitions in $\mathcal{DS}_2(t)$. Then $f(t) = 2^{\lfloor \frac{t}{2} \rfloor}$.*

Proof. Suppose that $t = 2k$, where k is a positive integer. Let $\lambda \in \mathcal{DS}_2(2k)$ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$, where $x_i > x_{i+1}$ for $i \in \mathbb{N}_{m-1}$. First, we show that the number of partitions in $\mathcal{DS}_2(2k)$ is equal to 2^k, i.e., $f(2k) = 2^k$. To this end, it suffices to show that for $k \geq 2$, the recurrence

$$f(2k) = f(2k-2) + 2f(2k-4) \tag{2.1}$$

holds, with initial values $f(0) = 1$ and $f(2) = 2$. Note that $2k-1$ and $2k+3$ can not simultaneously belong to $MD(\lambda)$, since $(2k-1)+(2k+3) = 2(2k+1)$. Thus, we divide the discussion into the following three cases:

(i) $2k-1,\ 2k+3 \notin MD(\lambda)$,

(ii) $2k-1 \in MD(\lambda),\ 2k+3 \notin MD(\lambda)$,

(iii) $2k-1 \notin MD(\lambda),\ 2k+3 \in MD(\lambda)$.

Here we only prove the assertion for Cases (i) and (ii). Case (iii) can be verified by similar arguments as (ii) and will be omitted.

Case (i). $2k-1, 2k+3 \notin MD(\lambda)$.
Let $\mathcal{W}(2k)$ be the set of such partitions λ, where $MD(\lambda) \subseteq \mathcal{H}^2_{2k}\backslash\{2k-1, 2k+3\}$. Define a map ϕ_{2k}: $\mathcal{W}(2k) \to \mathcal{DS}_2(2k-2)$ as follows. If $\lambda = \varnothing$, then $\phi_{2k}(\lambda) = \varnothing$. If $\lambda \neq \varnothing$, then we define $\phi_{2k}(\lambda)$ to be the partition μ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, where

$$y_i = \begin{cases} x_i & \text{if } x_i \in \mathcal{RH}^2_{2k}\backslash\{2k-1\}, \\ x_i - 4 & \text{if } x_i \in \mathcal{LH}^2_{2k}\backslash\{2k+3\}. \end{cases}$$

Then we deduce that $MD(\mu) \subseteq \mathcal{H}^2_{2k-2}$. It is easy to check that $MD(\mu)$ satisfies the three properties of Theorem 2.6. Hence $\mu \in \mathcal{DS}_2(2k-2)$.

For example, as shown in Fig. 4, if $\lambda = (10, 4, 3, 2, 1, 1, 1, 1, 1, 1) \in \mathcal{DS}_2(14)$, then $MD(\lambda) = \{19, 5, 1\}$, where $19 \in \mathcal{LH}^2_{14}\backslash\{17\}$ and $5, 1 \in \mathcal{RH}^2_{14}\backslash\{13\}$. Thus $\lambda \in \mathcal{W}(14)$. Applying the map ϕ_{14} to λ, we obtain $MD(\phi_{14}(\lambda)) = \{15, 5, 1\}$, that is, $\phi_{14}(\lambda) = (8, 4, 3, 2, 1, 1, 1, 1) \in \mathcal{DS}_2(12)$.

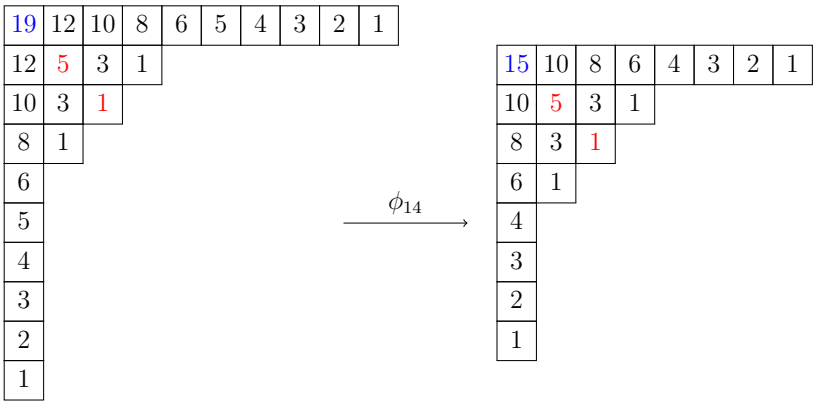


Fig. 4: An example of Case (i).

Conversely, given a partition $\mu \in \mathcal{DS}_2(2k-2)$ with $k \geq 2$, we can recover the partition $\lambda \in \mathcal{W}(2k)$ as follows. If $\mu = \varnothing$, then $\lambda = \varnothing$. If $\mu \neq \varnothing$ with $MD(\mu) =$

$\{y_1, y_2, \ldots, y_m\}$, then we define the partition λ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$, where

$$x_i = \begin{cases} y_i & \text{if } y_i \in \mathcal{RH}^2_{2k-2}, \\ y_i + 4 & \text{if } y_i \in \mathcal{LH}^2_{2k-2}. \end{cases}$$

It follows that $MD(\lambda) \subseteq \mathcal{H}^2_{2k}$ and $2k-1, 2k+3 \notin MD(\lambda)$. It is easy to check that $MD(\lambda)$ satisfies the three properties of Theorem 2.6. Hence $\lambda \in \mathcal{W}(2k)$. Therefore, the map ϕ_{2k} is reversible and a bijection. Thus, the number of such partitions λ in $\mathcal{W}(2k)$ is equal to $f(2k-2)$.

Case (ii). $2k-1 \in MD(\lambda), 2k+3 \notin MD(\lambda)$.
Let $\mathcal{U}(2k)$ denote the set of such partitions λ. By (ii) and (iii) of Theorem 2.6, we have $2k-3, 2k+5 \notin MD(\lambda)$ since $(2k-1)-(2k-3) = 2$ and $(2k-1)+(2k+5) = 2(2k+2)$. Let $\mathcal{U}'(2k)$ denote the set of partitions λ' obtained from λ by deleting the main diagonal hook of length $2k-1$, namely $MD(\lambda') = MD(\lambda)\backslash\{2k-1\} = \{x'_1, x'_2, \ldots, x'_{m-1}\}$, where $x'_i > x'_{i+1}$ for $i \in \mathbb{N}_{m-2}$. Then $MD(\lambda') \subseteq \mathcal{H}^2_{2k}\backslash\{2k-1, 2k-3, 2k+3, 2k+5\}$. Define a map φ_{2k}: $\mathcal{U}'(2k) \to \mathcal{DS}_2(2k-4)$ as follows. If $\lambda' = \varnothing$, that is, $\lambda = (2t-1)$, then $\varphi_{2k}(\lambda') = \varnothing$. If $\lambda' \neq \varnothing$, then we define $\varphi_{2k}(\lambda')$ to be the partition μ with $MD(\mu) = \{y_1, y_2, \ldots, y_{m-1}\}$, where

$$y_i = \begin{cases} x'_i & \text{if } x'_i \in \mathcal{RH}^2_{2k}\backslash\{2k-1, 2k-3\}, \\ x'_i - 8 & \text{if } x'_i \in \mathcal{LH}^2_{2k}\backslash\{2k+3, 2k+5\}. \end{cases}$$

Then we deduce that $MD(\mu) \subseteq \mathcal{H}^2_{2t-4}$. It is easy to check that $MD(\mu)$ satisfies the three properties of Theorem 2.6. This means that $\mu \in \mathcal{DS}_2(2k-4)$.

For example, as shown in Fig. 5, if $\lambda = (10, 7, 4, 3, 2, 2, 2, 1, 1, 1) \in \mathcal{DS}_2(12)$, then we have $MD(\lambda) = \{19, 11, 3\}$. Thus, $\lambda \in \mathcal{U}(12)$, $\lambda' = (10, 3, 2, 1, 1, 1, 1, 1, 1, 1) \in \mathcal{U}'(12)$, and $MD(\lambda') = \{19, 3\}$. Applying the map φ_{12} to λ', we obtain $MD(\varphi_{12}(\lambda')) = \{11, 3\}$, that is, $\varphi_{12}(\lambda') = (6, 3, 2, 1, 1, 1) \in \mathcal{DS}_2(8)$.

Conversely, given a partition $\mu \in \mathcal{DS}_2(2k-4)$, we can recover the partition $\lambda' \in \mathcal{U}'(2k)$. If $\mu = \varnothing$, then $\lambda' = \varnothing$. If $\mu \neq \varnothing$ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, then we define the partition λ' with $MD(\lambda') = \{x_1, x_2, \ldots, x_m\}$, where

$$x_i = \begin{cases} y_i & \text{if } x_i \in \mathcal{RH}^2_{2k-4}, \\ y_i + 8 & \text{if } x_i \in \mathcal{LH}^2_{2k-4}. \end{cases}$$

Then we derive $MD(\lambda') \in \mathcal{H}^2_{2k}$ and $2k-3, 2k-1, 2k+3, 2k+5 \notin MD(\lambda')$. It is easy to check that $MD(\lambda')$ satisfies the three properties of Theorem 2.6. This means that $\lambda' \in \mathcal{U}'(2k)$. Therefore, the map φ_{2k} is reversible and thus is a bijection. Furthermore, let λ be a partition with $MD(\lambda) = MD(\lambda') \cup \{2k-1\}$. Then $MD(\lambda)$ also satisfies the three properties of Theorem 2.6, implying $\lambda \in \mathcal{U}(2k)$, that is,

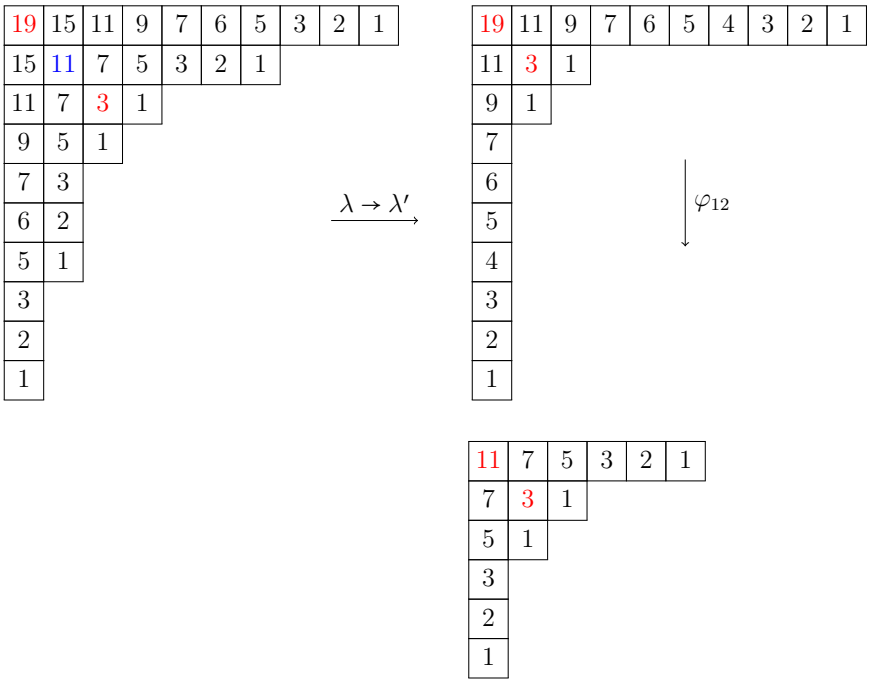


Fig. 5: An example of Case (ii).

$\lambda \in \mathcal{DS}_2(2k)$ with $2k-1 \in MD(\lambda)$. Therefore, the number of such partitions λ in $\mathcal{U}(2k)$ is equal to $f(2k-4)$.

Similarly, the number of partitions that satisfy Case (iii) is equal to $f(2k-4)$.

Combining the above three cases, we obtain (2.1). By the initial values $f(0)=1$ and $f(2)=2$, we derive that $f(2k)=2^k$. By Lemma 2.7, we further have $f(2k+1)=2^k$. Thus, we have $f(t)=2^{\lfloor \frac{t}{2} \rfloor}$ as desired. This completes the proof. ∎

Now we are ready to prove Theorem 1.2.

Proof of Theorem 1.2. Let $F(x)=\sum_{t\geq 0} f(t)x^t$. By Theorem 2.8 and the initial value $f(0)=1$, we obtain that

$$F(x)=\sum_{t\geq 0} f(t)x^t = 1+\sum_{t\geq 1} 2^{\lfloor \frac{t}{2} \rfloor}x^t = \sum_{t\geq 0} 2^t x^{2t} + \sum_{t\geq 0} 2^t x^{2t+1} = \frac{1+x}{1-2x^2}$$

as desired, and hence the proof is complete. ∎

2.2 Self-conjugate $(t, t+1, t+2, t+3)$-core partitions into distinct parts

By Theorem 2.5, it is easy to derive the following result.

Theorem 2.9 *Let t be a positive integer. Then λ is a partition in $\mathcal{DS}_3(t)$ if and only if $MD(\lambda)$ satisfies the following properties.*

(i) $MD(\lambda) \subseteq \mathcal{H}_t^3$.

(ii) *If $h_1, h_2 \in MD(\lambda)$ and $h_1 > h_2$, then $h_1 - h_2 \geq 4$.*

(iii) *If $h_1, h_2 \in MD(\lambda)$, then $h_1 + h_2$ is not equal to $2t$, $2t+2$, $2t+4$, or $2t+6$.*

In the subsequent analysis, we will derive a recurrence relation for the number $g(t)$ of partitions in $\mathcal{DS}_3(t)$.

Lemma 2.10 *Let $g(t)$ denote the number of partitions in $\mathcal{DS}_3(t)$. The number $g(t)$ satisfies the recurrence relation*

$$g(t) = g(t-1) + g(t-4) \tag{2.2}$$

for $t \geq 5$ with the initial values $g(0) = 1$, $g(1) = 1$, $g(2) = 2$, $g(3) = 2$, and $g(4) = 3$.

Proof. Let $t \geq 5$ and $\lambda \in \mathcal{DS}_3(t)$ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$, where $x_i > x_{i+1}$ for $i \in \mathbb{N}_{m-1}$. Then $MD(\lambda) \subseteq \mathcal{H}_t^3$. We divide the proof into two cases based on the parity of t. First, assume that t is even, and write $t = 2k$ with $k \geq 3$. We now distinguish two cases according to whether $t-1$ belongs to $MD(\lambda)$.

Case 1. $2k-1 \notin MD(\lambda)$.
Let $\Omega(2k)$ denote the set of partitions $\lambda \in \mathcal{DS}_3(2k)$ with $MD(\lambda) \subseteq \mathcal{H}_{2k}^3 \backslash \{2k-1\}$. Define a map $\tau_{2k} : \Omega(2k) \to \mathcal{DS}_3(2k-1)$ as follows. If $\lambda = \varnothing$, then $\tau_{2k}(\lambda) = \varnothing$. If $\lambda \neq \varnothing$, then we define $\tau_{2k}(\lambda)$ to be the partitions μ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, where

$$y_i = \begin{cases} x_i & \text{if } x_i \in \mathcal{RH}_{2k}^3 \backslash \{2k-1\}, \\ x_i - 2 & \text{if } x_i \in \mathcal{LH}_{2k}^3. \end{cases}$$

Thus, we have $MD(\mu) \subseteq \mathcal{H}_{2k-1}^3$. It is not difficult to verify that τ_{2k} is well-defined and $MD(\mu)$ satisfies the three properties of Theorem 2.9. This means that $\mu \in \mathcal{DS}_3(2k-1)$.

For example, if $t = 14$ and $\lambda = (14, 13, 12, 3, 3, 3, 3, 3, 3, 3, 3, 3, 2, 1) \in \mathcal{DS}_3(14)$, then we have $MD(\lambda) = \{27, 23, 19\} \in \mathcal{LH}_{14}^3$. Applying the map τ_{14} to λ, we obtain $MD(\tau_{14}(\lambda)) = \{25, 21, 17\}$, that is, $\tau_{14}(\lambda) = (13, 12, 11, 3, 3, 3, 3, 3, 3, 3, 3, 2, 1) \in \mathcal{DS}_3(13)$.

Conversely, given a partition $\mu \in \mathcal{DS}_3(2k-1)$, we can recover the partition $\lambda \in \Omega(2k)$. If $\mu = \varnothing$, then $\lambda = \varnothing \in \Omega(2k)$. If $\mu \neq \varnothing$ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$,

then we define the partition λ with $MD(\lambda) = \{x_1, x_2, \ldots, x_m\}$, where

$$x_i = \begin{cases} y_i & \text{if } y_i \in \mathcal{RH}^3_{2k-1}, \\ y_i + 2 & \text{if } y_i \in \mathcal{LH}^3_{2k-1}. \end{cases}$$

Thus, the construction of the map τ_{2k} is reversible and is a bijection. Therefore, the number of partitions of $\Omega(2k)$ equals $g(2k-1)$.

Case 2. $2k-1 \in MD(\lambda)$.
We deduce that $2k-3$, $2k+5$, and $2k+7$ do not belong to $MD(\lambda)$. Indeed, since

$$(2k-1)-(2k-3) = 2, (2k-1)+(2k+5) = 2(2k+2), \text{ and } (2k-1)+(2k+7) = 2(2k+3),$$

including any of these values would violate properties (ii) and (iii) of Theorem 2.9. Let $\Omega'(2k)$ denote the set of partitions λ' obtained from λ by removing the main diagonal hook of length $2k-1$. In this case, $MD(\lambda') = MD(\lambda)\backslash\{2k-1\} = \{x'_1, x'_2, \ldots, x'_{m-1}\}$, where $x'_i > x'_{i+1}$ for $i \in \mathbb{N}_{m-2}$. Then $MD(\lambda') \subseteq \mathcal{H}^3_{2k}\backslash\{2k-3, 2k-1, 2k+5, 2k+7\}$. Define a map $\vartheta_{2k} : \Omega'(2k) \to \mathcal{DS}_3(2k-4)$ as follows. If $\lambda' = \varnothing$, that is, $\lambda = (2k-1)$, then $\vartheta_{2k}(\lambda') = \varnothing$. If $\lambda' \neq \varnothing$, then we define $\vartheta_{2k}(\lambda')$ to be the partition μ with $MD(\mu) = \{y_1, y_2, \ldots, y_{m-1}\}$, where

$$y_i = \begin{cases} x'_i & \text{if } x'_i \in \mathcal{RH}^3_{2k}\backslash\{2k-3, 2k-1\}, \\ x'_i - 8 & \text{if } x'_i \in \mathcal{LH}^3_{2k}\backslash\{2k+5, 2k+7\}. \end{cases}$$

Thus, we have $MD(\mu) \subseteq \mathcal{H}^3_{2k-4}$. It is not difficult to verify that ϑ_{2k} is well-defined and $MD(\mu)$ satisfies the three properties of Theorem 2.9. This means that $\mu \in \mathcal{DS}_3(2k-4)$.

For example, if $t = 12$ and $\lambda = (11, 7, 3, 2, 2, 2, 2, 1, 1, 1, 1) \in \mathcal{DS}_3(12)$, then we have $MD(\lambda) = \{21, 11, 1\}$. It follows that $MD(\lambda') = \{21, 1\}$. Applying the map ϑ_{12} to λ, we obtain $MD(\vartheta_{12}(\lambda')) = \{13, 1\}$, that is, $\vartheta_{12}(\lambda') = (7, 2, 1, 1, 1, 1, 1) \in \mathcal{DS}_3(8)$.

Conversely, given a partition $\mu \in \mathcal{DS}_3(2k-4)$, we can recover the partition $\lambda' \in \Omega'(2k)$. If $\mu = \varnothing$, then $\lambda' = \varnothing \in \Omega'(2k)$. If $\mu \neq \varnothing$ with $MD(\mu) = \{y_1, y_2, \ldots, y_m\}$, then we define the partition λ' with $MD(\lambda') = \{x_1, x_2, \ldots, x_m\}$, where

$$x_i = \begin{cases} y_i & \text{if } y_i \in \mathcal{RH}^3_{2k-4}, \\ y_i + 8 & \text{if } y_i \in \mathcal{LH}^3_{2k-4}. \end{cases}$$

Thus, the construction of the map ϑ_{2k} is reversible and is a bijection. Therefore, the number of partitions of $\Omega'(2k)$ equals $g(2k-4)$.

Combining the above two cases, we obtain that

$$g(2k) = g(2k-1) + g(2k-4) \quad \text{for } k \geq 3. \tag{2.3}$$

Next, assume that t is odd. We need to consider two cases for the existence of $t+4$ in $MD(\lambda)$. These two cases can be handled similarly to the case of even t and are therefore omitted. Setting $t = 2k+1$ yields

$$g(2k+1) = g(2k) + g(2k-3) \quad \text{for } k \geq 2. \tag{2.4}$$

Combining (2.3) and (2.4), we obtain (2.2) as desired, completing the proof. ∎

Having proved Lemma 2.10, we are now ready to prove Theorem 1.3.

Proof of Theorem 1.3. Let $G(x) = \sum_{t\geq 0} g(t)x^t$. From the recurrence relation (2.2) and the initial values $g(0) = 1$, $g(1) = 1$, $g(2) = 2$, $g(3) = 2$, and $g(4) = 3$, we obtain that

$$G(x) = (1 + x + 2x^2 + 2x^3 + 3x^4) + x\big[G(x) - (1 + x + 2x^2 + 2x^3)\big] + x^4(G(x) - 1). \tag{2.5}$$

Then the formula (1.2) for $G(x)$ immediately follows from (2.5) as desired. This completes the proof. ∎

3 The largest size of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts

Recall that

$$d(\lambda) = (d_1, d_2, \ldots, d_{|MD(\lambda)|-1}),$$

where d_i is the difference between the i-th and the $(i+1)$-st main diagonal hook lengths of λ. For a partition $\lambda \in \mathcal{DS}_p(t)$ of largest size, we characterize $d(\lambda)$ in the following lemma.

Lemma 3.1 *For integers $t \geq 1$ and $p \geq 2$, let λ be a partition in $\mathcal{DS}_p(t)$ with the largest size. If $h_i, h_{i+1} \in MD(\lambda)$ for $1 \leq i \leq |MD(\lambda)| - 1$, then $d_i = 4$.*

Proof. Suppose that $\lambda \in \mathcal{DS}_p(t)$ achieves the largest size with $MD(\lambda) = \{h_1, h_2, \ldots, h_k\}$. We consider the following two cases.

Case 1. If $t - p \leq 1$, then $2t - 1 \leq t + p$. It follows that $\mathcal{H}_t^p = \mathcal{RH}_t^p$. Hence, by (ii) of Theorem 2.5, we have $d_i = 4$.

Case 2. If $t - p > 1$, then $2t - 1 > t + p$. It follows that $\mathcal{LR}_t^p \neq \varnothing$. Assume to the contrary that $MD(\lambda)$ contains elements h_j, h_{j+1} such that $d_j \geq 6$. Choose j as the smallest such integer. We then consider the following three subcases:

Subcase 2.1. Suppose that $h_j, h_{j+1} \in \mathcal{LH}_t^p$.

(1) Suppose that there does not exist $h_m \in MD(\lambda) \cap \mathcal{RH}_t^p$ such that $2(t+p+1)-h_j \leq h_m \leq 2t-2-h_{j+1}$. By the hypothesis, we have $2(t+r)-h_{j+1} \notin MD(\lambda)$ for each $r \in \mathbb{N}_{p-1}^0$. Then we can obtain a new partition μ, whose $MD(\mu)$ is obtained from $MD(\lambda)$ by replacing h_{j+1} with $h_{j+1}+2$.

(2) Suppose that there exists $h_m \in MD(\lambda) \cap \mathcal{RH}_t^p$ such that $2(t+p+1)-h_j \leq h_m \leq 2t-2-h_{j+1}$. Note that $2(t+p+1)-h_j \leq 2t-2-h_{j+1}$ must hold, otherwise such an h_m cannot exist. This implies that $d_j \geq 8$, $2(t+r)-h_m \in \mathcal{LH}_t^p$, and $2(t+r)-h_m \notin MD(\lambda)$ for each $r \in \mathbb{N}_p^0$. Choose h_m as the smallest such integer. Then we can obtain a new partition μ, whose $MD(\mu)$ is obtained from $MD(\lambda)$ by replacing h_m with h_j-4.

For example, let $\lambda \in \mathcal{DS}_5(13)$ with $MD(\lambda) = \{25, 19\}$. Then $d(\lambda) = (6)$ and $h_1 = 25, h_2 = 19 \in \mathcal{LH}_{13}^5$. By the above procedure in Subcase 2.1 (1), we obtain $\mu \in \mathcal{DS}_5(13)$ with $MD(\mu) = \{25, 21\}$.

Subcase 2.2. Suppose that $h_j \in \mathcal{LH}_t^p$ and $h_{j+1} \in \mathcal{RH}_t^p$. From (ii) of Theorem 2.6, we have $h_j < 2t-h_{j+1}$ or $h_j > 2t+2p-h_{j+1}$.

(1) Suppose that $h_j < 2t-h_{j+1}$. If $h_{j+1} = 2s-1 \in MD(\lambda) \cap \mathcal{RH}_t^p$ where $1 \leq s \leq p$, then we have $2t-2r-1 \in \mathcal{LH}_t^p$ and $2t-2r-1 \notin MD(\lambda)$ for each $r \in \mathbb{N}_p^0$. Otherwise, we have $2t+2r-h_{j+1} \in \mathcal{LH}_t^p$ and $2t+2r-h_{j+1} \notin MD(\lambda)$ for each $r \in \mathbb{N}_p^0$. By $p \geq 2$ and the choice of j, we can obtain a new partition μ, whose $MD(\mu)$ is obtained from $MD(\lambda)$ by replacing h_{j+1} with h_j+4.

(2) Suppose that $h_j > 2t+2p-h_{j+1}$. Let $\delta(S) = 1$ if the statement S is true and $\delta(S) = 0$ otherwise.

- If t is even and $h_{j+1} = t-1-2(s-1) \in MD(\lambda) \cap \mathcal{RH}_t^p$ where $1 \leq s \leq \lceil \frac{p}{2} \rceil$ (that is, h_{j+1} is the s-th largest element in $\mathcal{RH}_t^p$ for $1 \leq s \leq \lceil \frac{p}{2} \rceil$), then we have $t+p+1+2r+\delta(p \text{ is odd}) \in \mathcal{LH}_t^p$ and $t+p+1+2r+\delta(p \text{ is odd}) \notin MD(\lambda)$ for each $r \in \mathbb{N}_{\lfloor \frac{p}{2} \rfloor+s-1}^0$.
- If t is odd and $h_{j+1} = t-2-2(s-1) \in MD(\lambda) \cap \mathcal{RH}_t^p$ where $1 \leq s \leq \lfloor \frac{p}{2} \rfloor$ (that is, h_{j+1} is the s-th largest element in $\mathcal{RH}_t^p$ for $1 \leq s \leq \lfloor \frac{p}{2} \rfloor$), then we have $t+p+2+2r-\delta(p \text{ is odd}) \in \mathcal{LH}_t^p$ and $t+p+2+2r-\delta(p \text{ is odd}) \notin MD(\lambda)$ for each $r \in \mathbb{N}_{\lceil \frac{p}{2} \rceil+s-1}^0$.
- Otherwise, we have $2t+2r-h_{j+1} \in \mathcal{LH}_t^p$ and $2t+2r-h_{j+1} \notin MD(\lambda)$ for each $r \in \mathbb{N}_p^0$.

Then we can always obtain a new partition μ, whose corresponding $MD(\mu)$ is obtained from $MD(\lambda)$ by replacing h_{j+1} with h_j-4.

For example, let $\lambda \in \mathcal{DS}_4(23)$ with $MD(\lambda) = \{45, 41, 37, 21\}$. Then $d(\lambda) = (4, 4, 16)$ and $h_3 = 37 \in \mathcal{LH}^4_{23}$, $h_4 = 21 \in \mathcal{RH}^4_{23}$. So $h_3 = 37 > 2\times 23 + 2\times 4 - 21 = 33$. By the above procedure in Subcase 2.2 (2), we have $29, 31, 33 \notin MD(\lambda)$, then we obtain $\mu \in \mathcal{DS}_4(23)$ with $MD(\mu) = \{45, 41, 37, 33\}$.

Subcase 2.3. Suppose that $h_j, h_{j+1} \in \mathcal{RH}^p_t$.
By the choice of j, we have $MD(\lambda) \cap \mathcal{LH}^p_t = \varnothing$. Hence, we can obtain a new partition μ, whose $MD(\mu)$ is obtained from $MD(\lambda)$ by replacing h_{j+1} with $h_j - 4$.

The above analysis makes it easy to check that for each subcase, the $MD(\mu)$ we obtained always has a larger sum of elements than $MD(\lambda)$, which contradicts the assumption that λ has the largest size. This completes the proof. ∎

Remark 3.2 *Suppose that λ is a self-conjugate $(t, t+1, \ldots, t+p)$-core partition into distinct parts with the largest size. Let $h_i, h_{i+1} \in MD(\lambda)$. From Lemma 3.2 of [33], it follows that $4 \le d_i \le 6$ when $p = 1$. Based on Lemma 3.1, we obtain $d_i = 4$ when $p \ge 2$. The reason for this is that as p increases, the vertices in the lower part of the representation graph of $\mathcal{H}^p_t$ are connected by more lines, resulting in increasingly restrictive conditions on the selection of elements within $MD(\lambda)$, making the scenario of $d_i = 6$ impossible.*

Now we are ready to prove Theorem 1.5.

Proof of Theorem 1.5. For $p \ge 2$, suppose that $\lambda \in \mathcal{DS}_p(t)$ achieves the largest size. Let $MD(\lambda) = \{h_1, h_2, \ldots, h_k\}$ with $h_i > h_{i+1}$ for $i \in \mathbb{N}_{k-1}$. By Lemma 3.1, we have $d(\lambda) = (d_1, d_2, \ldots, d_{k-1})$, where $d_i = 4$ for all i. If $h_s \in \mathcal{LH}^p_t$ and $h_{s+1} \in \mathcal{RH}^p_t$, the structure of $\mathcal{RH}^p_t$ and $\mathcal{LH}^p_t$ implies that the difference $d_s = h_s - h_{s+1}$ is at least $2p+4$. Since $p \ge 2$, this gives $d_s \ge 8$, which is impossible. Therefore, we must have either $MD(\lambda) \subseteq \mathcal{LH}^p_t$ or $MD(\lambda) \subseteq \mathcal{RH}^p_t$.

Case 1. If $MD(\lambda) \subseteq \mathcal{RH}^p_t$, then λ has the largest size in $\mathcal{DS}_p(t)$ if and only if

$$MD(\lambda) = \begin{cases} \{t-1, t-5, \ldots, 7, 3\} & \text{if } t \equiv 0 \pmod 4, \\ \{t-2, t-6, \ldots, 7, 3\} & \text{if } t \equiv 1 \pmod 4, \\ \{t-1, t-5, \ldots, 5, 1\} & \text{if } t \equiv 2 \pmod 4, \\ \{t-2, t-6, \ldots, 5, 1\} & \text{if } t \equiv 3 \pmod 4. \end{cases}$$

By Lemma 2.2, the size $A(t)$ of λ is given by

$$A(t) = \begin{cases} \frac{t(t+2)}{8} & \text{if } t \text{ is even}, \\ \frac{(t-1)(t+1)}{8} & \text{if } t \text{ is odd}. \end{cases} \tag{3.1}$$

For example, if $t = 13$, $p = 12$, and λ is a partition in $\mathcal{DS}_{12}(13)$ with the largest size, then we have $MD(\lambda) = \{11, 7, 3\}$ as shown in Fig. 6, where we use red dots to represent the element of $MD(\lambda)$, thus $|\lambda| = 21$.



Fig. 6: The representation of $MD(\lambda) = \{11, 7, 3\}$.

Case 2. If $MD(\lambda) \subseteq \mathcal{LH}_t^p$ and $\mathcal{LH}_t^p \neq \varnothing$, then we have $t-p > 1$ and $2t-1 \in MD(\lambda)$. Thus, by Lemma 3.1, λ has the largest size in $\mathcal{DS}_p(t)$ if and only if

$$MD(\lambda) = \begin{cases} \{2t-1, 2t-5, \ldots, t+p+7, t+p+3\} & \text{if } t-p \equiv 0 \pmod 4, \\ \{2t-1, 2t-5, \ldots, t+p+8, t+p+4\} & \text{if } t-p \equiv 1 \pmod 4, \\ \{2t-1, 2t-5, \ldots, t+p+5, t+p+1\} & \text{if } t-p \equiv 2 \pmod 4, \\ \{2t-1, 2t-5, \ldots, t+p+6, t+p+2\} & \text{if } t-p \equiv 3 \pmod 4. \end{cases}$$

By Lemma 2.2, the size $B(t,p)$ of λ is given by

$$B(t,p) = \begin{cases} \frac{(3t+p+2)(t-p)}{8} & \text{if } t-p \equiv 0 \pmod 4, \\ \frac{(3t+p+3)(t-p-1)}{8} & \text{if } t-p \equiv 1 \pmod 4, \\ \frac{(3t+p)(t-p+2)}{8} & \text{if } t-p \equiv 2 \pmod 4, \\ \frac{(3t+p+1)(t-p+1)}{8} & \text{if } t-p \equiv 3 \pmod 4. \end{cases} \tag{3.2}$$

For example, if $t = 21$, $p = 3$, and λ is a partition in $\mathcal{DS}_3(21)$ with the largest size, then $t-p = 18 \equiv 2 \pmod 4$. Hence, we have $MD(\lambda) = \{41, 37, 33, 29, 25\}$ as shown in Fig. 7, where we use red dots to represent the element of $MD(\lambda)$, thus $|\lambda| = 165$.

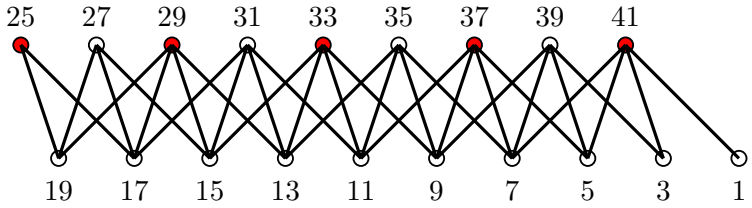


Fig. 7: The representation of $MD(\lambda) = \{41, 37, 33, 29, 25\}$.

Taken together, we find that the largest size of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts is equal to $\max\{A(t), B(t,p)\}$. This implies that the number of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts with the largest size is at most 2. This completes the proof. ∎

4 Conclusion and Future Work

Building on the definition of self-conjugate $(t, t+1)$-core partitions into distinct parts established in [33], we derived several formulas for the largest size of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts. For $p = 2$ and $p = 3$, we also obtain the number of such partitions.

However, a general pattern remains unclear for the number of such partitions for larger values of p, despite extensive numerical results on the number of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts from systematic calculations. This presents a significant open problem for future research.

Problem 4.1 *Study the enumeration of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts for $p \geq 4$.*

Table 3 lists the number of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts, covering all cases where $2 \leq t \leq 18$ and $p \leq 10$ (when $t = 1$, the number is always 1, corresponding to the empty set, regardless of p). Note that some of these sequences have appeared previously for $p = 4, 6, 8$ (see Sequences $A001590$, $A005314$ and $A340215$ of OEIS [27]), which provides supporting evidence for Problem 4.1.

Table 3: **The number of self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts for $2 \leq t \leq 18$ and $p \leq 10$.**

p	t																
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	2	3	4	6	9	13	19	28	41	60	88	129	189	277	406	595	872
2	2	2	4	4	8	8	16	16	32	32	64	64	128	128	256	256	512
3	2	2	3	4	6	8	11	15	21	29	40	55	76	105	145	200	276
4	2	2	3	3	6	6	11	11	20	20	37	37	68	68	125	125	230
5	2	2	3	3	5	6	9	11	16	20	28	36	50	65	89	117	159
6	2	2	3	3	5	5	9	9	16	16	28	28	49	49	86	86	151
7	2	2	3	3	5	5	8	9	14	16	24	28	41	49	70	85	120
8	2	2	3	3	5	5	8	8	14	14	24	24	41	41	70	70	119
9	2	2	3	3	5	5	8	8	13	14	22	24	37	41	62	70	104
10	2	2	3	3	5	5	8	8	13	13	22	22	37	37	62	62	104

In [33], the average size of the self-conjugate $(t, t+1)$-core partitions into distinct parts was calculated using their number and total size. For $p \geq 2$, the total size

of the self-conjugate $(t, t+1, \ldots, t+p)$-core partitions into distinct parts cannot be directly determined, making the average size difficult to compute. Thus we propose the following problem.

Problem 4.2 *Study the average size of the self-conjugate* $(t, t+1, \ldots, t+p)$*-core partitions into distinct parts for* $p \geq 2$.

Using Python, we obtained numerical results on the average size for $2 \leq t \leq 16$ and $p = 2, 3$, see Table 4, but no clear closed formula emerged. This remains a focus for future research.

Table 4: **The average size of self-conjugate** $(t, t+1, \ldots, t+p)$**-core partitions into distinct parts for** $2 \leq t \leq 16$ **and** $p = 2, 3$**.**

p	t														
	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
2	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{11}{4}$	$\frac{13}{4}$	$\frac{45}{8}$	$\frac{51}{8}$	$\frac{167}{16}$	$\frac{185}{16}$	$\frac{517}{32}$	$\frac{563}{32}$	$\frac{1503}{64}$	$\frac{1617}{64}$	$\frac{4093}{128}$	$\frac{4363}{128}$	$\frac{10727}{256}$
3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{4}{3}$	$\frac{13}{4}$	$\frac{13}{3}$	$\frac{51}{8}$	$\frac{82}{11}$	$\frac{161}{15}$	$\frac{265}{21}$	$\frac{478}{29}$	$\frac{149}{8}$	$\frac{1271}{55}$	$\frac{1973}{76}$	$\frac{3272}{105}$	$\frac{5009}{145}$

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